

GRADUATION PRESENTATION

RAG Pipeline Graduation Project

A focused exploration of Retrieval-Augmented Generation: From Python foundations through LangChain implementation to FastAPI deployment

 Graduation Project

 May 2026

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Core Modules

2

RAG Implementations

FastAPI

Project Stack

Python Foundations

🕒 Duration: 1 Month

01



Doctor Videos

Structured learning through professional video lectures



Form-based Assessments

Knowledge validation via structured examination forms



HackerRank Challenges

Practical coding exams on a competitive platform



Core Programming Skills

STEP 02

Resources Exploration



RAG Resource Research

🕒 Jan 1 - Feb 1

We Learned **Retrieval-Augmented Generation** architecture from 8 Researches.



AI as a Learning Tool

🕒 Feb 1 - Present

Using AI assistants for more **code understanding, readability** and facing errors.

⚠️ Subscription Challenge

Watched 25% of tutorial videos over 10 days before encountering paywall. Subscription required to access remaining content, limiting initial learning resources.

10 days to reach paywall

🔗 LangChain Framework

Key framework for building RAG applications, enabling integration of language models with external data sources.

Document Loaders

Vector Stores

Text Splitters

Chains

Agents

FRAMEWORK



RAG Implementation

➡ Powered by LangChain



Naive RAG

🕒 Mar 25 - Apr 13

Basic implementation using standard retrieval and generation chains.

Simple Retrieval

Direct Prompt

Baseline Model



Hardware Challenge

Encountered critical hardware compatibility issues during implementation that required extensive debugging and optimization.

Resolution: 3-7 days



Light RAG

🕒 May 6 - May 11

Optimized architecture with improved retrieval efficiency and context handling.

Optimized Retrieval

Context Windowing

Efficient

<> Light RAG Technical Stack

asyncio

numpy

lightrag

lightrag.utils

groq

langchain_huggingface

dotenv



Language Model

LLaMA 3.1 8B Instant

High-performance open-source LLM powering both Naive and Light RAG implementations with instant inference capabilities.

<> General Technical Stack

Core Libraries

shutil

ison

re

nest_asyncio

dotenv

groo

LangChain Components

document_loaders

text_splitters

huggingface

vectorstores

Evaluation Framework

test_case

base_model

faithfulness

answer_relevancy

API & RAG

fastapi

lightRag

RAG Deployment with FastAPI



RAG API Endpoints

Deploying **Naive RAG** and **Light RAG** as production-ready RESTful APIs.



RAG Evaluation Metrics

🕒 May 13 - May 19

Measuring retrieval accuracy, response quality, and latency for RAG performance optimization.



LangChain Integration

🕒 Mar 6 - Present

Bridging LangChain RAG pipelines with FastAPI for scalable **RAG-as-a-Service**.



FastAPI

RAG

LangChain

Evaluation

RAG Journey Complete



Python Foundation

Established core programming skills essential for RAG development.



LangChain Mastery

Deep understanding of RAG architecture and implementation tools.



RAG Implementation

Successfully built both Naive RAG and Light RAG systems.



FastAPI Deployment

Deployed RAG solutions with comprehensive evaluation metrics.



Thank You

RAG Pipeline Graduation Project

Questions & Discussion